0.1 A robust relation $\delta\Pi \longrightarrow \delta\Pi^-$

We established that :

$$\partial^{\mathbf{n}}(\delta\Pi - d\Gamma_x^*\Pi_x)(x) = 0 \quad \text{for } |\mathbf{n}| < 2+s. \tag{1}$$

We will see how this specifies uniquely the coefficients $\mathrm{d}\Gamma_x^*$. Indeed, by definition,

$$\mathrm{d}\Gamma_x^* := \sum_{|\mathbf{n}|<2+s} \mathrm{d}\pi_x^{(n)}\Gamma_x^*\partial_{z_{\mathbf{n}}}.$$

And as

$$\partial_{z_{\mathbf{n}}} z_{\mathbf{m}} = \delta_{\mathbf{n},\mathbf{m}},$$

It follows that

$$\forall \mathbf{m}, |\mathbf{m}| \geq 2+s, \mathrm{d}\Gamma_x^* z_m = 0.$$

Additionnaly, $\mathrm{d}\Gamma_x^*$ behaves like a derivation :

$$\begin{cases} \mathrm{d}\Gamma_x^*\pi\pi' = (\mathrm{d}\Gamma_x^*\pi)(\Gamma_x^*\pi') + (\Gamma_x^*\pi)(\mathrm{d}\Gamma_x^*\pi'), \\ \mathrm{d}\Gamma_x^* 1 = 0, \\ \mathrm{d}\Gamma_x^*\mathbf{z}_3 = 0. \end{cases}$$

This follows directly from the behaviour of $\partial_{\mathbf{z}_n}$ which is a derivation, and Γ_x^* which is an algebra morphism. As a consequence, together with its finiteness property, $\mathrm{d}\Gamma_x^*$ is determined by its values on the finitely many coordinates $\{\mathbf{z}_n\}_{|\mathbf{n}|<2+s}$.

Knowing $\delta\Pi$ and (1), we actually know $\mathrm{d}\Gamma_x^*\partial^{\mathbf{n}}\Pi_x(x)$ for all $\mathbf{n}$ such that $|\mathbf{n}| < 2+s$. We also know this fact about the jet:

$$\frac{1}{\mathbf{n}!}\partial^{\mathbf{n}}\Pi_x(x) - \mathbf{z}_{\mathbf{n}} \in \mathbb{R}[[\mathbf{z}_3, \{\mathbf{z}_{\mathbf{m}}\}_{|\mathbf{m}|<|\mathbf{n}|}]]. \tag{2}$$

Now, an induction argument shows that we thus know $\mathrm{d}\Gamma_x^*\mathbf{z}_n$ for all $\mathbf{n}$ such that $|\mathbf{n}| < 2+s$.

Proof. The base case is is clear because $\Pi_x(x) - \mathbf{z_0} \in \mathbb{R}[[\mathbf{z}_3]]$ and $\mathrm{d}\Gamma_x^*\mathbf{z}_3 = 0$ so

$$\mathrm{d}\Gamma_x^*\mathbf{z_0} = \mathrm{d}\Gamma_x^*\Pi_x(x).$$

Now let $k \in \mathbb{N}$, $k < 2+s$ be such that we know $\mathrm{d}\Gamma_x^*\mathbf{z_m}$ for all $\mathbf{m}$ such that $|\mathbf{m}| < k$. For all $\mathbf{n}$ with $|\mathbf{n}| = k$, (2) shows that $\mathrm{d}\Gamma_x^*\mathbf{z_n}$ is a function of $\mathrm{d}\Gamma_x^*\partial^{\mathbf{n}}\Pi_x(x)$ and some $\mathrm{d}\Gamma_x^*\mathbf{z_m}$ with $|\mathbf{m}| < k$, which are known by hypothesis. □

This will be useful to explicit a robust relation $\delta\Pi \longrightarrow \delta\Pi^-$. By robust we mean that the map doesn't rely on diverging constant such as c, like if we used the first definition of $\delta\Pi^-$:

$$\delta\Pi^- := (3\mathbf{z}_3\Pi^2 + c)\delta\Pi + \delta\xi_\rho\mathbf{1}.$$

We claim that :

$$(\delta\Pi^- - \delta\xi_\rho\mathbf{1} - \mathrm{d}\Gamma_x^*\Pi_x^-)(x) = 0 \tag{3}$$

Proof. Let's work out $\mathrm{d}\Gamma_x^*\Pi_x^-(x)$:

$$\begin{aligned}
\mathrm{d}\Gamma_x^*\Pi_x^-(x) &= \mathrm{d}\Gamma_x^*(\mathbf{z}_3\Pi_x^3 + c\Pi_x + \xi_\rho\mathbf{1})(x) \\
&= ((\Gamma_x^*\mathbf{z}_3)(\mathrm{d}\Gamma_x^*\Pi_x^3) + (\Gamma_x^*c)(\mathrm{d}\Gamma_x^*\Pi_x) + \mathrm{d}\Gamma_x^*\xi_\rho\mathbf{1})(x) \quad \text{(derivation properties)} \\
&= (3\mathbf{z}_3(\mathrm{d}\Gamma_x^*\Pi_x)(\Gamma_x^*\Pi_x^2) + c\,\mathrm{d}\Gamma_x^*\Pi_x)(x) \quad \text{(same properties, and } \Gamma_x^*c = c, \Gamma_x^*\mathbf{z}_3 = \mathbf{z}_3) \\
&= (3\mathbf{z}_3\Pi^2 + c)(\mathrm{d}\Gamma_x^*\Pi_x)(x) \quad \text{(because } \Gamma_x^*\Pi_x = \Pi) \\
&= (3\mathbf{z}_3\Pi^2 + c)(\delta\Pi)(x) \quad \text{(by (2) for } \mathbf{n} = \mathbf{0}) \\
&= (\delta\Pi^- - \delta\xi_\rho)(x) \quad \text{(by definition of } \delta\Pi^-)
\end{aligned}$$

□

The calculation is easier in this direction altough we are truely going in the opposite direction. The success of our approach may be due to the fact that we use an approximation of $\delta\Pi$ at each x with the germ $\mathrm{d}\Gamma_x^*\Pi_x$, as opposed to only utilizing $\Pi_0; \Pi_0^-$.

Equation (3) will give us the qualitative result we need to quantitatively reconstruct $\delta\Pi^-$ from the germ $\mathrm{d}\Gamma_x^*\Pi_x^-$ (see section 3.2 of [BOT25]). We also see the main induction argument coming : from $\delta\Pi$ we get this relation, and we will integrate the estimate we obtain on $\delta\Pi^-$ to get an estimate on $\delta\Pi$.

0.2 Main result

The goal is to prove convergence of the model in the forme of the following estimates :

$$\mathbb{E}^{\frac{1}{p}}|\Pi_{x\beta r}(x)|^p \lesssim r^{|\beta|}, \tag{4}$$
$$\mathbb{E}^{\frac{1}{p}}|\Pi^-_{x\beta r}(x)|^p \lesssim r^{|\beta|-2}, \tag{5}$$
$$\mathbb{E}^{\frac{1}{p}}|(\Gamma^*_x)^\gamma_\beta|^p \lesssim r^{|\beta|-|\gamma|} \quad \text{for all populated } \gamma. \tag{6}$$

The first two estimates are homogeneity estimates on the germs Π_x and Π^-_x, and together with the last one, we also get a coherence estimate on these two germs.

$$\mathbb{E}^{\frac{1}{p}}|(\Pi_{x+y\beta} - \Pi_{x\beta})_r(x)|^p \lesssim r^\alpha(r+|y|)^{|\beta|-\alpha}, \tag{7}$$
$$\mathbb{E}^{\frac{1}{p}}|\Pi^-_{\beta r}(x)|^p \lesssim r^{3\alpha}(r+|x|)^{|\beta|-2-3\alpha}, \quad \text{provided } \beta \neq 0. \tag{8}$$

This estimate gives us useful insight : the increments $\Pi_{x+y\beta} - \Pi_{x\beta}$ have local regularity α (which is negative) at an arbitrary point, but vanishes at the order $|\beta|$ when y goes to 0. Finally, at infinity, they grow at order $|\beta| - \alpha$.

Proof. Proof of (7) We know that $\Pi_{x+y\beta} - \Pi_{x\beta} = (\Gamma^*_y - id)\Pi_{x\beta} = \sum_\gamma (\Gamma^*_y - id)^\gamma_\beta \Pi_{x\gamma}$. We apply the mollification operator $(\cdot)_r$ and evaluate at x. Using the triangle inequality with respect to the norm $\mathbb{E}^{\frac{1}{p}}|\cdot|^p$, and then Hölder's inequality, we get :

$$\mathbb{E}^{\frac{1}{p}}|(\Pi_{x+y\beta} - \Pi_{x\beta})_r(x)|^p \lesssim \sum_\gamma \mathbb{E}^{\frac{1}{2p}}|(\Gamma^*_x - id)^\beta_\gamma|^{2p}\mathbb{E}^{\frac{1}{2p}}|\Pi_{x\gamma r}(x)|^{2p}. \tag{9}$$

Now appealing to $|\gamma| \geq |\beta| \implies (\Gamma^*_x - id)^\gamma_\beta = 0$ to restrict the sum and get rid of id, and estimates (4) and (6), we get :

$$\mathbb{E}^{\frac{1}{p}}|(\Pi_{x+y\beta} - \Pi_{x\beta})_r(x)|^p \lesssim \sum_{|\gamma|\leq|\beta|} |y|^{|\beta|-|\gamma|} r^{|\gamma|}. \tag{10}$$

We also know that $|\gamma| \geq \alpha$ for all populated γ (see (62) in [BOT25], α is the roughest regularity of the model, it is the regularity of Π_0), so we can restrict the sum to $|\gamma| \geq \alpha$. Factorizing by r^α, and using a binomial expansion, we get (7). □

The proof for Π^- is analog.

We are not trying to apply a reconstruction theorem here, but coherence and homogeneity are the two relevant metrics to measure germs of distributions. In order to prove these estimates, we will first prove similar estimates on $\delta\Pi$ and $\delta\Pi^-$, which are the Malliavin derivatives of Π and Π^- with respect to the noise ξ, or rather on their germs $\mathrm{d}\Gamma^*_x\Pi_x$ and $\mathrm{d}\Gamma^*_x\Pi^-_x$ (cf section 2.6 from [BOT25] to see why these are the right germs). We will take advantage of the fact that $\delta\Pi$ and $\delta\Pi^-$ are related by a robust relation, and that they are more regular, by a order $\frac{D}{2}$.

The shape of the estimate will be slighty different as we are going to take a local L^2-norm in x instead of an L^∞ norm. This is dictated by what we are able to prove for the base case : a pointwise estimate is to strong, and we use an L^2-norm because of the L^2 nature of $\dot{H}_s$ ($\ni \delta\Pi^-_0 = \delta\xi_\rho$). The proof of the base case relies on Fourier transform. Finally, we cannot expect square integrability of $\mathbb{E}^{\frac{2}{q}}|(\delta\Pi - \mathrm{d}\Gamma^*_x\Pi_x)_{\beta r}(x)|^q$ at infinity, so we take a local norm.

We will thus try to prove the following estimates :

$$(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(\mathrm{d}\Gamma^*_{x+y}\Pi^-_{x+y} - \mathrm{d}\Gamma^*_x\Pi^-_x)_{\beta r}(x+y)|^q)^{\frac{1}{2}} \lesssim r^{2\alpha}(r+|y|)^{\alpha+\frac{D}{2}}(|y|+r+R)^{|\beta|-2-3\alpha} \quad \text{(coherence)} \tag{11}$$

$$(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(\mathrm{d}\Gamma^*_x)^\gamma_\beta|^q)^{\frac{1}{2}} \lesssim R^{\frac{D}{2}+\beta-\gamma} \quad \text{for populated } \gamma \neq \mathrm{pp} \quad \text{(homogeneity)} \tag{12}$$

This will allow us to prove by reconstruction that :

$$(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(\delta\Pi^- - \mathrm{d}\Gamma^*_x\Pi^-_x)_{\beta r}(x)|^q)^{\frac{1}{2}} \lesssim r^{3\alpha+\frac{D}{2}}(r+R)^{|\beta|-2-3\alpha} \quad \text{(reconstruction)} \tag{13}$$

And deduce a reconstruction bound for $\delta\Pi$ by integration (see below) :

$$(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(\delta\Pi - \mathrm{d}\Gamma^*_x\Pi_x)_{\beta r}(x)|^q)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-\alpha} \quad \text{(reconstruction)} \tag{14}$$

What is cructial here is the exponent $3\alpha + \frac{D}{2}$, in the coherence estimate of $\mathrm{d}\Gamma^*_x\Pi^-_x$ and in the reconstruction bound for $\delta\Pi^-$. It indeed plays the role of γ in the reconstruction theorem presented in [BL23], and is positive by assumption in [BOT25]. Because of this, the reconstuction theorem ensures uniqueness of the reconstruction $\delta\Pi^-$. The same goes for $\delta\Pi$ with the exponent $\alpha + \frac{D}{2}$ which is even more positive as $\alpha < 0$. This cascades down to show that our model is uniquely caracterized.

Actually we can't directly integrate (13) because of terms of to high homogeneity in $\mathrm{d}\Gamma^*_x\Pi^-_x$. We will need to split this sum into two parts :

$$(\mathrm{d}\Gamma^*_x\Pi^-_x)_\beta = \sum_{|\gamma|\geq 2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi^-_{x\gamma} + \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi^-_{x\gamma}$$

The first part is the one we will be able to control by homogeneity, because of the lower bound on $|\gamma|$:

$$(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |\sum_{|\gamma|\geq 2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi^-_{x\gamma r}(x)|^q)^{\frac{1}{2}} \lesssim r^{\alpha-2+\frac{D}{2}}(r+R)^{|\beta|-\alpha} \quad \text{(homogeneity)} \tag{15}$$

we thus get by triangle inequality and (13):

$$(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(\delta\Pi^-_\beta - \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi^-_{x\gamma})_r(x)|^q)^{\frac{1}{2}} \lesssim r^{\alpha-2+\frac{D}{2}}(r+R)^{|\beta|-\alpha}, \tag{16}$$

which we are able to integrateto find :

$$(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |(\delta\Pi_\beta - \sum_{|\gamma|<2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi_{x\gamma})_r(x)|^q)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-\alpha}. \tag{17}$$

Because we also have the homogeneity estimate of the high homogeneity part of $\mathrm{d}\Gamma^*_x\Pi_x$:

$$(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}} |\sum_{|\gamma|\geq 2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \Pi_{x\gamma r}(x)|^q)^{\frac{1}{2}} \lesssim r^{\alpha+\frac{D}{2}}(r+R)^{|\beta|-\alpha} \quad \text{(homogeneity)}, \tag{18}$$

we are able to prove (14) by triangle inequality.

Let us prove the homogeneity estimate (15), but first try the direct approach to prove the homogeneity bound of the germ $\mathrm{d}\Gamma^*_x \delta\Pi^-_x$ without splitting, to see the difference in exponents. We will end up using both computation anyway.

$$\begin{aligned}
\Big(\int_{B_R} \mathrm{d}x \, \mathbb{E}^{\frac{2}{q}} \big| (\mathrm{d}\Gamma^*_x \Pi^-_x)_\beta \big|^q \Big)^{\frac{1}{2}} &= \Big(\int_{B_R} \mathrm{d}x \, \mathbb{E}^{\frac{2}{q}} \Big| \sum_\gamma (\mathrm{d}\Gamma^*_x)^\gamma_\beta \, \Pi^-_{x\gamma r}(x) \Big|^q \Big)^{\frac{1}{2}} \\
&\lesssim \sum_\gamma \Big(\int_{B_R} \mathrm{d}x \, \mathbb{E}^{\frac{2}{2q}} \big| (\mathrm{d}\Gamma^*_x)^\gamma_\beta \big|^{2q} \mathbb{E}^{\frac{2}{2q}} \big| \Pi^-_{x\gamma r}(x) \big|^{2q} \Big)^{\frac{1}{2}} \quad \text{(Hölder)} \\
&\lesssim \sum_{\substack{\gamma \neq \mathrm{pp} \\ |\gamma|_\prec < |\beta|_\prec}} \Big(\int_{B_R} \mathrm{d}x \, \mathbb{E}^{\frac{2}{2q}} \big| (\mathrm{d}\Gamma^*_x)^\gamma_\beta \big|^{2q} \Big)^{\frac{1}{2}} \sup_{x \in B_R} \mathbb{E}^{\frac{1}{2q}} \big| \Pi^-_{x\gamma r}(x) \big|^{2q} \\
&\lesssim \sum_{\alpha \le |\gamma| < \frac{D}{2} + |\beta|} R^{\frac{D}{2} + |\beta| - |\gamma|} r^{|\gamma| - 2} \quad \text{(by induction hypothesis)} \\
&\lesssim r^{\alpha - 2} \sum_{\alpha \le |\gamma| < \frac{D}{2} + |\beta|} R^{\frac{D}{2} + |\beta| - |\gamma|} r^{|\gamma| - \alpha} \\
&\lesssim r^{\alpha - 2} (R + r)^{\frac{D}{2} + |\beta| - \alpha}
\end{aligned}$$

The condition $\gamma \neq \mathrm{pp}$ comes from the fact that $\gamma = \mathrm{pp} \implies \Pi^-_{x\gamma} = 0$. The condition $|\gamma|_\prec < |\beta|_\prec$ comes from the stric triangularity of $\mathrm{d}\Gamma^*_x$ with respect to $|\cdot|_\prec$. Finally, we always have $|\gamma| \ge \alpha$, and $|\gamma| < |\beta| + \frac{D}{2}$ also comes from population conditions on $\mathrm{d}\Gamma^*_x$.

Now, say that we truncate the sum from below by $|\gamma| \ge 2 + s = \alpha + \frac{D}{2}$, and try to prove (15) :

$$\begin{aligned}
\Big(\int_{B_R} \mathrm{d}x \, \mathbb{E}^{\frac{2}{q}} \Big| \sum_{|\gamma| \ge 2+s} (\mathrm{d}\Gamma^*_x)^\gamma_\beta \, \Pi^-_{x\gamma r}(x) \Big|^q \Big)^{\frac{1}{2}} &\lesssim \sum_{\alpha + \frac{D}{2} \le |\gamma| < \frac{D}{2} + |\beta|} R^{\frac{D}{2} + |\beta| - |\gamma|} r^{|\gamma| - 2} \\
&\lesssim r^{\alpha + \frac{D}{2} - 2} \sum_{\alpha + \frac{D}{2} \le |\gamma| < \frac{D}{2} + |\beta|} R^{\frac{D}{2} + |\beta| - |\gamma|} r^{|\gamma| - \alpha - \frac{D}{2}} \\
&\lesssim r^{\alpha + \frac{D}{2} - 2} (R + r)^{|\beta| - \alpha}
\end{aligned}$$

which is better, and matches the reconstruction bound (13). So this allow us to not degrade our exponents when using the triangle inequality to get (16), which in turn allows us to get the good reconstruction bound on $\delta\Pi$ (14).

0.3 Reconstruction of $\mathrm{d}\Gamma_x^*\Pi_x^-$

Proof of (13). For convenience, denote $F_x \coloneqq (\mathrm{d}\Gamma_x^*\Pi_x^-)_\beta$ and introduce the diagonal evaluation for this germ : $(EF)(x) = F_x(x)$. We are going to prove,

$$\Big(\int_{B_R} \mathrm{d}x \mathbb{E}^{\frac{2}{q}}|(EF_\tau - F_{x\tau})_{T-\tau}(x)|^q\Big)^{\frac{1}{2}} \lesssim (\sqrt[4]{T})^{3\alpha+\frac{D}{2}}(\sqrt[4]{T}+R)^{|\beta|-2-3\alpha}, \tag{19}$$

because from (3), we have the qualitative : $\delta\Pi_\beta^-(x) = \lim_{\tau\to 0}(\mathrm{d}\Gamma_x^*\Pi_x^-)_{\beta\tau}(x) = \lim_{\tau\to 0} EF_\tau(x)$, so taking the limit $\tau \to 0$, we hope to recover the targeted estimate.

Let $n \in \mathbb{N}$ and $\tau = 2^{-n}T$. Then, we can write

$$\begin{aligned}(EF_\tau - F_{x\tau})_{T-\tau}(x) &= (EF_\tau)_{T-\tau} - (EF_{2\tau})_{T-2\tau}\\ &+ (EF_{2\tau})_{T-2\tau} - (EF_{4\tau})_{T-2\tau}\\ &+ (EF_{4\tau})_{T-4\tau} - (EF_{8\tau})_{T-8\tau} + \dots\\ &+ (EF_{\frac{T}{2}})_{\frac{T}{2}} - EF_T(x)\\ &= \sum_{k=0}^{n-1}(EF_{2^{k-n}T})_{(1-2^{k-n})T} - (EF_{2^{k+1-n}T})_{(1-2^{k+1-n})T}\end{aligned}$$

Let $t = 2^{k-n}T$, we then have to estimate $(EF_t)_{T-t} - (EF_{2t})_{T-2t} = ((EF_t)_t - EF_{2t})_{T-2t}$ (semi-group property).

$$\begin{aligned}((EF_t)_t - EF_{2t})_{T-2t}(x) &= \int \mathrm{d}y\, \psi_{T-2t}(y)((EF_t)_t - EF_{2t})(x-y)\\ &= \int \mathrm{d}y\, \psi_{T-2t}(y)((EF_t)_t - F_{x-y2t})(x-y)\\ &= \int \mathrm{d}y\, \psi_{T-2t}(y)((EF_t) - F_{x-yt})_t(x-y)\\ &= \int \mathrm{d}y\, \psi_{T-2t}(y)\int \mathrm{d}y'\psi_t(y')((EF_t) - F_{x-yt})(x-y-y')\\ &= \int \mathrm{d}y\, \psi_{T-2t}(y)\int \mathrm{d}y'\psi_t(y')(F_{x-y-y't} - F_{x-yt})(x-y-y')\end{aligned}$$

We see the increments of the germ $\mathrm{d}\Gamma_x^*\Pi_x^-$ appearing. Using the fact that $\int_{B_R} \mathrm{d}x f^2(x+y) \leq \int_{B_{R+|y|}} \mathrm{d}x' f^2(x')$, we have

$$\Big(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}}|(F_{x-y-y't}-F_{x-yt})(x-y-y')|^q\Big)^{\frac{1}{2}} \lesssim \Big(\int_{B_{R+|y|}} \mathrm{d}x\mathbb{E}^{\frac{2}{q}}|(F_{x-y't}-F_{xt})(x-y')|^q\Big)^{\frac{1}{2}} \tag{20}$$

And the r.h.s is estimated by (11) with $(\sqrt[4]{t}, -y', R+|y|)$ playing the role of (r, y, R).

Thus, by triangle inequality, we have :

$$\begin{aligned}&\Big(\int_{B_R} \mathrm{d}x\mathbb{E}^{\frac{2}{q}}|((EF_t)_t - EF_{2t})_{T-2t}(x)|^q\Big)^{\frac{1}{2}}\\ &\lesssim (\sqrt[4]{t})^{2\alpha}\int \mathrm{d}y\,|\psi_{T-2t}(y)|\int \mathrm{d}y'|\psi_t(y')|(|y'|+\sqrt[4]{t})^{\alpha+\frac{D}{2}}(|y'|+\sqrt[4]{t}+R+|y|)^{|\beta|-2-3\alpha}\end{aligned} \tag{21}$$

Let $\mu = \sqrt[4]{t}$ and $\lambda = \sqrt[4]{T-2t}$. By change of variable,

$$\begin{aligned}&(\sqrt[4]{t})^{2\alpha}\int dy\,|\Psi_{T-2t}(y)|\int dy'\,|\Psi_t(y')|\,(|y'|+\sqrt[4]{t})^{\alpha+\frac{D}{2}}(|y'|+\sqrt[4]{t}+R+|y|)^{|\beta|-2-3\alpha}\\ &= (\sqrt[4]{t})^{2\alpha}\iint |\Psi_1(z)|\,|\Psi_1(z')|\left[\mu^{\alpha+\frac{D}{2}}(1+|z'|)^{\alpha+\frac{D}{2}}\right](\mu|z'|+\lambda|z|+\mu+R)^{|\beta|-2-3\alpha}\,dz\,dz'\end{aligned} \tag{22}$$

We separate scale variables from integration variables :

$$\mu|z'|+\lambda|z|+\mu+R\leq(\mu+\lambda+R)(1+|z|+|z'|) \tag{23}$$

We obtain :

$$\begin{aligned}\text{l. h. s. of } (22) &\leq (\sqrt[4]{t})^{2\alpha}\cdot\mu^{\alpha+\frac{D}{2}}\cdot(\mu+\lambda+R)^{|\beta|-2-3\alpha}\\ &\quad\times\iint|\Psi_1(z)|\,|\Psi_1(z')|\,(1+|z'|)^{\alpha+\frac{D}{2}}(1+|z|+|z'|)^{|\beta|-2-3\alpha}\,dz\,dz'\end{aligned} \tag{24}$$

Ψ_1 is a Schwartz function, so the integral is a macroscopic constant. Replacing μ and λ by their value, and because $\sqrt[4]{t}+\sqrt[4]{T-2t}\lesssim\sqrt[4]{T}$, we get :

$$\Big(\int_{B_R}\mathrm{d}x\mathbb{E}^{\frac{2}{q}}|((EF_t)_t-EF_{2t})_{T-2t}(x)|^q\Big)^{\frac{1}{2}}\lesssim(\sqrt[4]{t})^{3\alpha+\frac{D}{2}}(\sqrt[4]{T}+R)^{|\beta|-2-3\alpha} \tag{25}$$

Recall that $t=2^{k-n}T$, for $k=0,\dots,n-1$. Plugging in the telescoping sum, we get :

$$\begin{aligned}\Big(\int_{B_R}\mathrm{d}x\mathbb{E}^{\frac{2}{q}}|(EF_\tau-F_{x\tau})_{T-\tau}(x)|^q\Big)^{\frac{1}{2}} &\lesssim(\sqrt[4]{T}+R)^{|\beta|-2-3\alpha}\sum_{k=0}^{n-1}(\sqrt[4]{2^{(k-n)}T})^{3\alpha+\frac{D}{2}}\\ &\lesssim(\sqrt[4]{T})^{3\alpha+\frac{D}{2}}(\sqrt[4]{T}+R)^{|\beta|-2-3\alpha}\sum_{i=1}^{n}\rho^i\quad(\rho:=(\sqrt[4]{2})^{-3\alpha-\frac{D}{2}})\\ &\lesssim(\sqrt[4]{T})^{3\alpha+\frac{D}{2}}(\sqrt[4]{T}+R)^{|\beta|-2-3\alpha}\end{aligned}$$

where the final inequality is justified by the fact that $\rho<1$ because of assumption $3\alpha+\frac{D}{2}>0$.

Now, we need to prove that taking $\tau\to0$, we get (13). By the semigroup property, we have :

$$(EF_\tau-F_{x\tau})_{T-\tau}(x)=(EF_\tau)_{T-\tau}(x)-(\mathrm{d}\Gamma_x^*\Pi_x^-)_{\beta T}(x).$$

Let's prove $(EF_\tau)_{T-\tau}(x)\longrightarrow\delta\Pi_T^-(x)$ for all $x\in B_R$. By definition :

$$(EF_\tau)_{T-\tau}(x)=\int\mathrm{d}y\,\psi_{T-\tau}(y)\,(\mathrm{d}\Gamma_{x-y}^*\Pi_{x-y}^-)_{\beta\tau}(x-y). \tag{26}$$

By (3) and the pointwise convergence $\psi_{T-\tau}\to\psi_T$, the integrand converges a.s. to $\psi_T(y)\,\delta\Pi_\beta^-(x-y)$ as $\tau\to0$. To apply the dominated convergence theorem, we claim that $(\mathrm{d}\Gamma_z^*\Pi_z^-)_\beta$ is of polynomial growth in z. Since ψ is a Schwartz function and therefore decays faster than any polynomial, this provides an integrable dominating function. Applying Fatou's lemma successively in the probability space and then in B_R, the uniform bound (19) passes to the limit and yields (13). □

Remarque 0.1. To see that $(\mathrm{d}\Gamma_z^*\Pi_z^-)_\beta$ has polynomial growth, recall that we work with a fixed $\rho>0$. So all the objects constructed from ξ_ρ - which is of polynomial growth - by convolutions, and polynomial operations, are of polynomial growth.

A more pedestrian argument consists in relying on induction hypothesis :

$$(\mathrm{d}\Gamma_z^*\Pi_z^-)_\beta=\sum_{\substack{\gamma\neq\mathrm{pp}\\|\gamma|_\prec<|\beta|_\prec}}(\mathrm{d}\Gamma_z^*)_\beta^\gamma\Pi_{z\gamma}^-.$$

First, we prove a.s. polynomial growth of $(\mathrm{d}\Gamma_x^*)_\beta^\gamma$, $\gamma\neq$pp. By (12) with $R=2^k$:

$$\int_{B_{2^k}}\mathbb{E}^{\frac{2}{q}}|(\mathrm{d}\Gamma_x^*)_\beta^\gamma(x)|^q\,\mathrm{d}x\lesssim2^{k(D+2(|\beta|-|\gamma|))}.$$

Set $Z_k := \left(\int_{B_{2^k}} |(\mathrm{d}\Gamma_x^*)_\beta^\gamma(x)|^q \,\mathrm{d}x \right)^{2/q}$. By Markov's inequality, for any $\varepsilon > 0$:

$$\mathbb{P}\Big(Z_k > 2^{k(D+2(|\beta|-|\gamma|)+\varepsilon)}\Big) \leq \frac{\mathbb{E}[Z_k]}{2^{k(D+2(|\beta|-|\gamma|)+\varepsilon)}} \lesssim 2^{-k\varepsilon}.$$

This is summable in k, so by the Borel-Cantelli lemma, a.s. for all k large enough:

$$\int_{B_{2^k}} |(\mathrm{d}\Gamma_x^*)_\beta^\gamma(x)|^q \,\mathrm{d}x \lesssim_\omega 2^{kN_\varepsilon},$$

for some $N_\varepsilon > 0$. That is, $(\mathrm{d}\Gamma_x^*)_\beta^\gamma$ has a.s. polynomial L^q-growth in x. Because $(\mathrm{d}\Gamma_x^*)_\beta^\gamma$ is smooth (we work at a fixed $\rho > 0$), Sobolev'inequality gives us almost sure polynomial growth uniformly in x.

Secondly by induction hypothesis, (5) holds for $|\gamma|_\prec < |\beta|_\prec$.

Bibliography

[BL23] Lucas Broux and David Lee. Besov reconstruction. *Potential Analysis*, 59(4):1875–1912, 12 2023.

[BOT25] Lucas Broux, Felix Otto, and Markus Tempelmayr. Lecture notes on Malliavin calculus in regularity structures. *Stochastics and Partial Differential Equations: Analysis and Computations*, 09 2025.